

Homological Algebra Lecture Notes

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1 Lecture 1 (October 1st 2026)

Preview: We basically spend some time doing category theory, and the theory of functors with some attention paid to R -Modules. There is some additional mastery material that we need to cover as well.

Basics of Category Theory

Definition 1.1. A *category* $\mathcal{C}$ consists of

1. a collection of ‘objects’ called $\text{ob}(\mathcal{C})$
2. $\forall A, B \in \text{ob}(\mathcal{C})$ we have a set of “morphisms” $\text{Hom}_{\mathcal{C}}(A, B)$
3. An identity (or a unit) for each object $\text{id}_A \in \text{Hom}_{\mathcal{C}}(A, A)$
4. A composition rule for A, B, C which is: $\text{Hom}_{\mathcal{C}}(B, C) \times \text{Hom}_{\mathcal{C}}(A, B) \rightarrow \text{Hom}_{\mathcal{C}}(A, C)$ It is simple to think of $g \circ f \in \text{Hom}_{\mathcal{C}}(A, C)$ where $g \in \text{Hom}_{\mathcal{C}}(B, C)$ and $f \in \text{Hom}_{\mathcal{C}}(A, B)$ so that we really have composition of functions here.
5. These functions are also associative such that for $f \in \text{Hom}_{\mathcal{C}}(A, B), g \in \text{Hom}_{\mathcal{C}}(B, C), h \in \text{Hom}_{\mathcal{C}}(C, D)$ we have $h \circ (g \circ f) = (h \circ g) \circ f$ and the same for units $\text{id}_B \circ f = f \circ \text{id}_A = f$.

So what’s cool about this is that this is basically a group without the inverses requirement. It becomes hard to rigorously define what this collection of objects really is, and especially to decide if it is a set, so they handwaved this, but we’ll come back to it later here. Now, we continue with an example.

Example 1.2. Placeholder